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Semiconductor device and semiconductor device manufacturing method

Abstract

A semiconductor device includes: a substrate; a first semiconductor chip; a first adhesive layer; a second semiconductor chip; a second adhesive layer; and a spacer. The substrate has a first surface. The first semiconductor chip is provided above the first surface. The first adhesive layer is provided on a lower surface, which is opposed to the substrate, of the first semiconductor chip and contains a plurality of types of resins different in molecular weight. The second semiconductor chip is provided between the substrate and the first adhesive layer. The second adhesive layer covers surroundings of the second semiconductor chip in a view from a normal direction of a first surface, and contains at least one type of the resin lower in molecular weight than the other resins among the plurality of types of resins contained in the first adhesive layer. The spacer covers surroundings of the second adhesive layer in the view from the normal direction of the first surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application is based upon and claims the benefit of priority from No. 2021-090138, filed May 28, 2021, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate generally to a semiconductor device and a semiconductor device manufacturing method.

BACKGROUND

(3) A structure in which, for example, memory chips are stacked on a controller chip flip-chip connected to a substrate is known in semiconductor device packaging processes. However, it has been often difficult to appropriately provide the memory chips on the substrate depending on, for example, a chip size relationship between the controller chip and the memory chips.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view illustrating an example of configurations of a semiconductor device according to a first embodiment.

(2) FIG. 2 is a cross-sectional view illustrating an example of a semiconductor device manufacturing method according to the first embodiment.

(3) FIG. 3 is a cross-sectional view, subsequent to FIG. 2, illustrating the example of the semiconductor device manufacturing method.

(4) FIG. 4 is a cross-sectional view, subsequent to FIG. 3, illustrating the example of the semiconductor device manufacturing method.

(5) FIG. 5 is a cross-sectional view, subsequent to FIG. 4, illustrating the example of the semiconductor device manufacturing method.

(6) FIG. 6 is a cross-sectional view, subsequent to FIG. 5, illustrating the example of the semiconductor device manufacturing method.

(7) FIG. 7 is a cross-sectional view, subsequent to FIG. 6, illustrating the example of the semiconductor device manufacturing method.

(8) FIG. 8 is a cross-sectional view, subsequent to FIG. 7, illustrating the example of the semiconductor device manufacturing method.

(9) FIG. 9 is a top view illustrating an example of configurations of through-holes of a spacer and surroundings of the through-holes according to the first embodiment.

(10) FIG. 10 is a top view illustrating an example of configurations of the semiconductor wafer before dicing and surroundings of the semiconductor wafer according to the first embodiment.

(11) FIG. 11 is a cross-sectional view illustrating an example of a semiconductor device manufacturing method according to a second embodiment.

(12) FIG. 12 is a cross-sectional view, subsequent to FIG. 11, illustrating the example of the semiconductor device manufacturing method.

(13) FIG. 13 is a cross-sectional view, subsequent to FIG. 12, illustrating the example of the semiconductor device manufacturing method.

(14) FIG. 14 is a cross-sectional view, subsequent to FIG. 13, illustrating the example of the semiconductor device manufacturing method.

DETAILED DESCRIPTION

(15) Embodiments provide a semiconductor device and a semiconductor device manufacturing method capable of providing chips on a substrate more appropriately.

(16) In general, according to at least one embodiment, a semiconductor device includes: a substrate; a first semiconductor chip; a first adhesive layer; a second semiconductor chip; a second

adhesive layer; and a spacer. The substrate has a first surface. The first semiconductor chip is provided above the first surface. The first adhesive layer is provided on a lower surface, which is opposed to the substrate, of the first semiconductor chip and contains a plurality of types of resins different in molecular weight. The second semiconductor chip is provided between the substrate and the first adhesive layer. The second adhesive layer covers surroundings of the second semiconductor chip in a view from a normal direction of a first surface, and contains at least one type of the resin lower in molecular weight than the other resins among the plurality of types of resins contained in the first adhesive layer. The spacer covers surroundings of the second adhesive layer in the view from the normal direction of the first surface.

(17) Embodiments of the disclosure will be described hereinafter with reference to the drawings. It is to be noted that the present embodiments are not intended to limit the disclosure. In the following embodiments, a vertical direction of an interconnection substrate indicates a relative direction when a surface on which a semiconductor chip is provided is assumed as an upper surface and often differs from a vertical direction according to a gravitational acceleration. The drawings are either schematic or conceptual and proportions and the like of elements are not necessarily identical to actual proportions and the like. In the specification and the drawings, similar elements to those described previously with respect to the drawings already referred to are denoted by the same reference signs and detailed descriptions thereof are omitted as appropriate.

First Embodiment

(18) FIG. 1 is a cross-sectional view illustrating an example of configurations of a semiconductor device **1** according to a first embodiment. The semiconductor device **1** includes an interconnection substrate **10**, semiconductor chips **20** and **30** to **33**, adhesive layers **40** to **43**, a spacer **50**, an adhesive layer **51**, an adhesive layer **60**, a metal material **70**, a resin layer **80**, a bonding wire **90**, and an encapsulation resin **91**. The semiconductor device **1** is, for example, a NAND flash memory package.

(19) The interconnection substrate **10** may be a printed circuit board or an interposer including interconnection layers **11** and an insulating layer **15**. For example, any of low resistance metals such as copper, nickel, and alloys thereof is used for the interconnection layers **11**. For example, an insulating material such as a glass epoxy resin is used for the insulating layer **15**. In FIG. 1, the interconnection layers **11** are provided only on a front surface and a rear surface of the insulating layer **15**, respectively. Alternatively, the interconnection substrate **10** may have a multilayer interconnection structure in which a plurality of interconnection layers **11** and a plurality of insulating layers **15** are stacked. The interconnection substrate **10** may have, for example, a through via **12** penetrating a front surface and a rear surface of the interconnection substrate **10** as in the interposer.

(20) A solder resist layer **14** provided on the interconnection layer **11** is provided on the front surface of the interconnection substrate **10**. The solder resist layer **14** is an insulating layer that protects the interconnection layer **11** from the metal material **70** and that prevents a short circuit defect.

(21) The solder resist layer **14** provided on the interconnection layer **11** is also provided on the rear surface of the interconnection substrate **10**. Metal bumps **13** are provided on the interconnection layer **11** exposed from the solder resist layer **14**. The metal bumps **13** are provided for electrically connecting other components, not illustrated, to the interconnection substrate **10**.

(22) The semiconductor chip **20** is provided on a front surface-side, i.e., an upper surface-side of the interconnection substrate **10**. The semiconductor chip **20** is provided between the interconnection substrate **10** and the adhesive layer **40**. The semiconductor chip **20** is, for example, a controller chip that controls the memory chips. A semiconductor element, not illustrated, is provided on a surface, which faces to the interconnection substrate **10**, of the semiconductor chip **20**. The semiconductor element may be, for example, a CMOS (Complementary Metal Oxide Semiconductor) circuit configuring a controller. Electrode pillars **21** electrically connected to the

semiconductor element are provided on the rear surface (lower surface) of the semiconductor chip **20**. For example, any of low resistance metals such as copper, nickel, and alloys thereof is used for the electrode pillars **21**.

(23) A metal material **70** surrounds each electrode pillar **21** that serves as a connection bump. The electrode pillar **21** is electrically connected to the interconnection layer **11** exposed in an opening of the solder resist layer **14** via the metal material **70**. For example, any of low resistance metals such as a solder, silver, and copper is used for the metal material **70**. The metal material **70** covers, for example, part of the interconnection layer **11** of the interconnection substrate **10** in the opening and also covers part of side surfaces of the electrode pillar **21** of the semiconductor chip **20**. The metal material **70** thereby electrically connects the electrode pillar **21** of the semiconductor chip **20** to the interconnection layer **11** of the interconnection substrate **10**.

(24) A resin layer (underfill) **80** surrounds the metal material **70** and between the semiconductor chip **20** and the interconnection substrate **10**. The resin layer **80**, which is formed from, for example, a hardened NCP (Non-Conductive Paste), covers and protects surroundings of the semiconductor chip **20**.

(25) More specifically, in a view from a normal direction of a surface **F1** of the interconnection substrate **10**, the semiconductor chip **20** is disposed in such a manner that the overall semiconductor chip **20** overlaps the semiconductor chip **30**. That is, a chip size of the semiconductor chip **20** is smaller than a chip size of the semiconductor chip **30**.

(26) The semiconductor chip **30** adheres onto the semiconductor chip **20** via the adhesive layer **40**. Therefore, the semiconductor chip **30** is provided above the upper surface, i.e., surface **F1** of the interconnection substrate **10**. The semiconductor chip **30** is, for example, a memory chip including a NAND flash memory. The semiconductor chip **30** has semiconductor elements, not illustrated, on a front surface, i.e., upper surface thereof. The semiconductor elements may be, for example, a memory cell array and a peripheral circuit (CMOS circuit) of the memory cell array. The memory cell array may be a stereoscopic memory cell array on which a plurality of memory cells are disposed three-dimensionally. In addition, the semiconductor chip **31** adheres onto the semiconductor chip **30** via the adhesive layer **41**. The semiconductor chip **32** adheres onto the semiconductor chip **31** via the adhesive layer **42**. The semiconductor chip **33** adheres onto the semiconductor chip **32** via the adhesive layer **43**. The semiconductor chips **31** to **33**, like the semiconductor chip **30**, are each, for example, a memory chip including a NAND flash memory. The semiconductor chips **30** to **33** may be identical memory chips. In FIG. 1, the semiconductor chips **30** to **33** serving as four memory chips are stacked as well as the semiconductor chip **20** serving as the controller chip. Nevertheless, the number of stacked semiconductor chips may be equal to or smaller than three or may be equal to or greater than five.

(27) The spacer **50** covers surroundings of the adhesive layer **60** in the view from the normal direction of the surface **F1** of the interconnection substrate **10**. The spacer **50** adheres onto the front surface, i.e., upper surface of the interconnection substrate **10** via the adhesive layer **51**. The adhesive layer **51** is provided between the interconnection substrate **10** and the spacer **50**. In addition, the spacer **50** has a through-hole **50h**. The semiconductor chip **20** and the adhesive layer **60** are disposed in the through-hole **50h**. A width of the through-hole **50h** in the view from the normal direction of the surface **F1** of the interconnection substrate **10** is slightly larger than a width of the semiconductor chip **20**, as will be described later.

(28) The adhesive layer **60** covers surroundings of the semiconductor chip **20** in the view from the normal direction of the surface **F1** of the interconnection substrate **10**. The adhesive layer **60** is disposed in such a manner that the overall adhesive layer **60** overlaps the adhesive layer **40** in the view from the normal direction of the surface **F1** of the interconnection substrate **10**. This is because the adhesive layer **60** is formed by flow off of part of the adhesive layer **40** downward, as will be described later. The adhesive layer **60** covers side surfaces of the semiconductor chip **20** in the through-hole **50h** of the spacer **50**. That is, the adhesive layer **60** buries a gap between a

combination of the semiconductor chip **20** and the resin layer **80**, and the spacer **50**. It is noted that detailed configurations of the adhesive layer **40**, the spacer **50**, and the adhesive layer **60** will be described later.

(29) The bonding wire **90** is connected to the interconnection substrate **10** and freely selected pads of the semiconductor chips **30** to **33**. To connect the semiconductor chips **30** to **33** by the bonding wire **90**, the semiconductor chips **30** to **33** are stacked to be staggered each by as much as a pad. It is noted that the semiconductor chip **20** is not wire bonded since being flip-chip connected by the electrode pillars **21**. It is noted that the semiconductor chip **20** may be not only flip-chip connected by the electrode pillars **21** but also wire bonded.

(30) Furthermore, the encapsulation resin **91** encapsulates the semiconductor chips **20** and **30** to **33**, the adhesive layers **40** to **43** and **51**, the spacer **50**, the bonding wire **90**, and the like. The semiconductor device **1** is thereby configured as one semiconductor package including the plurality of semiconductor chips **20** and **30** to **33** on the interconnection substrate **10**.

(31) Detailed internal configurations of the adhesive layer **40**, the spacer **50**, and the adhesive layer **60** will next be described.

(32) The adhesive layer **40** is provided on a lower surface, i.e., surface **30b**, which is opposed to the interconnection substrate **10**, of the semiconductor chip **30**. The adhesive layer **40** contains a plurality of types of resins different in molecular weight. The adhesive layer **40** contains, for example, a high-molecular-weight base material and a low-molecular-weight epoxy resin. It is noted that the low-molecular-weight epoxy resin may be a resin at a relatively low softening point or, to be specific, a resin lower in softening point than the base material. The adhesive layer **40** preferably contains the low-molecular-weight epoxy resin at a relatively high compounding ratio. A material for the adhesive layer **40** may be the same as a material for the adhesive layers **41** to **43**. The adhesive layer **40** also contains a filler, not illustrated. In addition, when the adhesive layer **40** is thin, an amount of the resin flowing off from the adhesive layer **40** is small and it is possibly difficult to bury the through-hole **50h** with the adhesive layer **60**. Preferably, the adhesive layer **40** has a thickness enough to be able to bury at least the through-hole **50h**. It is noted that the necessary thickness of the adhesive layer **40** possibly varies depending on the material or the like for the adhesive layer **40**. The thickness of the adhesive layer **40** is, for example, equal to or larger than half a thickness of the semiconductor chip **20** or half a thickness of a combination of the spacer **50** and the adhesive layer **51**.

(33) The adhesive layer **60** contains at least one type of the resin lower in molecular weight than the other resins among the plurality of types of resins contained in the adhesive layer **40**. The adhesive layer **60** contains, for example, the low-molecular-weight epoxy resin. Furthermore, a filler concentration of the adhesive layer **60** is lower than a filler concentration of the adhesive layer **40**. More specifically, the adhesive layer **60** hardly contains a filler. This is because it is difficult for the high-molecular-weight base material and the high-molecular-weight filler, e.g., silica filler to flow off from the adhesive layer **40** and the base material and the filler, e.g., silica filler remain in the adhesive layer **40**.

(34) The spacer **50** supports the semiconductor chip **30** and the adhesive layer **40**. In addition, the spacer **50** is relatively high in rigidity. More specifically, the spacer **50** is higher in rigidity than the adhesive layer **40**. This enables the spacer **50** to keep a shape even when, for example, being pressurized from a semiconductor chip **30**-side at a time of mounting the semiconductor chip **30**. The spacer **50** is, for example, a resin layer. Since the resin layer is easier to work, it is possible to facilitate forming the through-hole **50h** in which the semiconductor chip **20** is accommodated. A material for the spacer **50** is, for example, polyimide.

(35) Furthermore, a height of an upper surface, i.e., surface **50a**, which is opposed to the adhesive layer **40**, of the spacer **50** from the surface **F1** is smaller than a height of an upper surface, i.e., surface **20a**, which is opposed to the adhesive layer **40**, of the semiconductor chip **20** from the surface **F1**. Therefore, the height of the spacer **50** is set in such a manner that part of the

semiconductor chip **20** sinks into the adhesive layer **40**, as will be described later.

(36) A method of manufacturing the semiconductor device **1** will next be described.

(37) FIGS. **2** to **8** are cross-sectional views illustrating an example of the method of manufacturing the semiconductor device **1** according to the first embodiment. FIGS. **2** to **5** illustrate processes before a semiconductor wafer **W** is diced. FIGS. **6** to **8** illustrate processes of dicing the semiconductor wafer **W** into semiconductor chips and mounting each semiconductor chip **30** on the interconnection substrate **10**.

(38) FIG. **9** is a top view illustrating an example of configurations of the through-holes **50h** of the spacer **50** and surroundings of the through-holes **50h** according to the first embodiment. FIG. **10** is a top view illustrating an example of configurations of the semiconductor wafer **W** before dicing and surroundings of the semiconductor wafer **W** according to the first embodiment. FIGS. **2** to **8** are also enlarged cross-sectional views of one through-hole **50h** and the surroundings of the through-hole **50h** illustrated in FIGS. **9** and **10**. Line A-A of FIGS. **9** and **10** is a line along which FIGS. **2** to **8** that are cross-sectional views are taken.

(39) In the example illustrated in FIGS. **9** and **10**, the semiconductor wafer **W**, the adhesive layer **40**, the spacer **50**, and a dicing tape **DT** are stacked almost concentrically. As illustrated in FIG. **10**, a size of the spacer **50** is larger than a size of the adhesive layer **40**. In addition, the size of the adhesive layer **40** is larger than a size of the semiconductor wafer **W**.

(40) In FIGS. **9** and **10**, regions **R1** and **R2** are regions above the spacer **50**. The region **R2** is the region where the through-holes **50h** are formed, i.e., the region where the adhesive layer **40** and the semiconductor wafer **W** are provided. It is noted that disposition and the like of the through-holes **50h** illustrated in FIGS. **9** and **10** are given by way of example. The region **R1** is the region different from the region **R2** and outside of the region **R2**. The region **R1** is the region outside of the adhesive layer **40** and the semiconductor wafer **W**. Marks **50m** are provided in the region **R1**. The marks **50m** are wafer lamination marks. The marks **50m** are used to position the semiconductor wafer **W** so that positions of semiconductor elements formed on the semiconductor wafer **W** correspond to positions of the through-holes **50h**.

(41) First, as illustrated in FIG. **2**, the spacer **50** is made to adhere onto the dicing tape **DT** via the adhesive layer **51**. For example, the spacer **50** having the adhesive layer **51** provided in advance is provided on the dicing tape **DT**.

(42) Next, as illustrated in FIG. **3**, each through-hole **50h** is formed in the spacer **50**. The through-hole **50h** is formed to penetrate the dicing tape **DT**, the adhesive layer **51**, and the spacer **50**. The through-hole **50h** is formed by, for example, punching part of the dicing tape **DT**, the adhesive layer **51**, and the spacer **50** by a mold from a vertical direction of FIG. **3**.

(43) It is noted that the position of each through-hole **50h** is set in response to the position of each semiconductor element formed on the semiconductor wafer **W**. That is, the position of the through-hole **50h** is set in response to the position of each semiconductor chip **30** before dicing.

(44) More specifically, before formation of each through-hole **50h**, the two marks **50m** are formed in the region **R1** illustrated in FIG. **9**. The marks **50m** are, for example, through-holes. Subsequently, a plurality of through-holes **50h** are formed in the region **R2** with reference to positions of the two marks **50m**. That is, the through-holes **50h** are formed at the positions with reference to the positions of the two marks **50m**.

(45) Next, as illustrated in FIG. **4**, the adhesive layer **40** is provided on the spacer **50**. Since the adhesive layer **40** is provided after forming the through-holes **50h**, the through-holes **50h** are not provided in the adhesive layer **40**.

(46) Next, as illustrated in FIG. **5**, the semiconductor wafer **W** is provided on the adhesive layer **40**. More specifically, the semiconductor wafer **W** is provided on the adhesive layer **40** in response to the positions of the plurality of through-holes **50h**. The semiconductor elements of the semiconductor chips **30** are provided on an upper surface of the semiconductor wafer **W**. A lower surface of the semiconductor wafer **W** is subjected to backgrinding so that the semiconductor wafer

W has a desired thickness.

(47) When the adhesive layer **40** is disposed herein on the spacer **50**, visibility to the through-holes **50h** decreases. The semiconductor wafer W is, therefore, provided in the region R2 with reference to the positions of the marks **50m** disposed in the region R1 illustrated in FIG. **10**. That is, the marks **50m** are disposed apart from the through-holes **50h**, so that it is possible to determine an appropriate position of mounting the semiconductor wafer W on the basis of the marks **50m**.

(48) Next, as illustrated in FIG. **6**, the semiconductor wafer W is diced into semiconductor chips **30**. The semiconductor wafer W is diced using, for example, dicing blades DB. The adhesive layer **40**, the spacer **50**, and the adhesive layer **51** are also diced together with the semiconductor wafer W. The semiconductor wafer W is cut into a chip size along a dicing line DL illustrated in FIG. **10**. It is noted that FIG. **10** illustrates part of the dicing line DL.

(49) Next, as illustrated in FIG. **7**, the semiconductor chip **30** is made to adhere onto the interconnection substrate **10** via the adhesive layer **51**. In FIG. **7**, the semiconductor chip **20** is mounted on the surface F1 of the interconnection substrate **10** in advance. Therefore, the adhesive layer **51**, the spacer **50**, the adhesive layer **40**, and the semiconductor chip **30** are provided on the surface F1 so that the semiconductor chip **20** is located in the through-hole **50h**.

(50) Moreover, in FIG. **7**, the interconnection substrate **10** and the semiconductor chip **20** are heated. The interconnection substrate **10** and the semiconductor chip **20** are heated using, for example, a heater, not illustrated, provided below the interconnection substrate **10**.

(51) Furthermore, a thickness T of the combination of the spacer **50** and the adhesive layer **51** is smaller than the thickness of the semiconductor chip **20**. The upper surface, i.e., surface **20a** of the semiconductor chip **20** thereby comes in contact with the adhesive layer **40** before the adhesive layer **51** comes in contact with the interconnection substrate **10**. This facilitates transferring heat from the heater provided on an interconnection substrate **10**-side to the adhesive layer **40**.

(52) Next, as illustrated in FIG. **8**, the adhesive layer **60** is formed to bury the through-hole **50h**. The semiconductor chip **30** obtained by dicing is mounted on the interconnection substrate **10** at a predetermined pressure. The adhesive layer **40** and the spacer **50** are thereby pressurized. In addition, the heat from the heater provided on the interconnection substrate **10**-side is transferred to the adhesive layer **40** via the interconnection substrate **10** and the semiconductor chip **20**.

(53) When the adhesive layer **40** is heated, part of the resins contained in the adhesive layer **40** are melted. The low-molecule-weight component resin among the resins in the adhesive layer **40** tends to be melted by the heat. The melted resin flows off (distills) from the adhesive layer **40** and buries the through-hole **50h** in response to the pressure during mounting. That is, by heating and pressurizing the adhesive layer **40**, the adhesive layer **60** is formed in the through-hole **50h**. It is noted that a main component of the adhesive layer **60** is the low-molecular-weight epoxy resin since it is difficult for the high-molecular-weight base material and the high-molecular-weight filler to flow off from the adhesive layer **40**.

(54) The interconnection substrate **10** is heated at a temperature, for example, equal to or lower than 200° C. The interconnection substrate **10** may be heated at a temperature, for example, equal to or lower than 150° C. depending on the material for the adhesive layer **40**.

(55) Furthermore, as illustrated in FIG. **8**, the height from the surface F1 of the interconnection substrate **10** to the surface **50a** of the spacer **50** is smaller than the height from the surface F1 of the interconnection substrate **10** to the surface **20a** of the semiconductor chip **20**. In the process illustrated in FIG. **8**, it is thereby possible that the upper surface of the semiconductor chip **20** presses the adhesive layer **40** to facilitate transferring the heat to the adhesive layer **40**. As a result, the adhesive layer **60** can flow off from the adhesive layer **40** easily. Furthermore, it is possible to reduce a possibility of generating a gap between the upper surface of the semiconductor chip **20** and the adhesive layer **40** and, therefore, further facilitates burying the through-hole **50h** with the adhesive layer **60**. It is noted that the spacer **50** is less prone to deformation even receiving the heat due to the high rigidity and the height of the spacer **50** can be maintained.

(56) After the process illustrated in FIG. 8, the semiconductor chips **31** to **33**, the adhesive layers **41** to **43**, and the bonding wire **90** are provided. Subsequently, the elements such as the semiconductor chips **20** and **31** to **33** are encapsulated with the encapsulation resin **91**, thereby completing the semiconductor device **1** illustrated in FIG. 1.

(57) As described so far, according to the first embodiment, the adhesive layer **60** flows off from the adhesive layer **40** to bury the through-hole **50h** of the spacer **50**. The adhesive layer **60**, therefore, surrounds the semiconductor chip **20** disposed in the through-hole **50h**. It is thereby possible to bury the semiconductor chip **20** with the adhesive layer **60** more appropriately and mount (dispose) the semiconductor chip **30** on the interconnection substrate **10** and the semiconductor chip **20** more appropriately.

(58) Furthermore, in the view from the normal direction of the surface **F1** of the interconnection substrate **10**, the width of the through-hole **50h** is larger than the width of the semiconductor chip **20**. If the width of the through-hole **50h** is nearly the same as that of the semiconductor chip **20**, the spacer **50** possibly interferes with the semiconductor chip **20** in the process illustrated in FIG. 7. If the width of the through-hole **50h** is sufficiently larger than that of the semiconductor chip **20**, a gap widens between the semiconductor chip **20** and the spacer **50** in the through-hole **50h**. In this case, the through-hole **50h** (gap) may be difficult to bury with the adhesive layer **60**. It is, therefore, preferable that the width of the through-hole **50h** is slightly larger than the width of the semiconductor chip **20**. The width of the through-hole **50h** is set larger than the width of the semiconductor chip **20** in consideration of, for example, position accuracy. The position accuracy includes, for example, position accuracy for the through-hole **50h** and position accuracy at the time of mounting the semiconductor chips **20** and **30**. The width of the through-hole **50h** is set, for example, larger than the width of the semiconductor chip **20** by a predetermined value, e.g., 100 μm .

(59) As a comparison, there is a structure of burying the semiconductor chip **20** with a thick adhesive layer, e.g., DAF (Die Attach Film) provided on the lower surface of the semiconductor chip **30**. With this structure, however, a region of an adhesive portion between the adhesive layer and the interconnection substrate **10** shrinks when, for example, the size of the semiconductor chip **30** is relatively small or when the size of the semiconductor chip **30** is closer to the size of the semiconductor chip **30**. As the region of the adhesive portion is smaller, the semiconductor chip **30** becomes more prone to being inclined and the adhesive layer becomes more prone to being peeled off from the interconnection substrate **10** during curing. Furthermore, a thickness of the adhesive layer with which the semiconductor chip **20** is buried increases since the semiconductor chip **30** is difficult to be thinned and a height of connection with the interconnection substrate **10** is necessary when the semiconductor chip **20** is flip-chip connected. That is, when the size of the semiconductor chip **30** is relatively small, the structure of burying the semiconductor chip **20** with the thick adhesive layer is difficult to adopt.

(60) According to the first embodiment, by contrast, the spacer **50** surrounds the semiconductor chip **20**. The semiconductor chip **30** becomes prone to warp as being thinner. The spacer **50** supports the semiconductor chip **30** so that the semiconductor chip **30** is made flat. In addition, since the spacer **50** supports the semiconductor chip **30**, even the thinner semiconductor chip **30** is capable of preventing occurrence of a warp or the like. Furthermore, since the spacer **50** supports the semiconductor chip **30**, it is possible to prevent an inclination of the semiconductor chip **30**. Moreover, because of no difference in height by the semiconductor chip **20**, a lower surface of the spacer **50** adheres onto the interconnection substrate **10** via the adhesive layer **51** appropriately. It is thereby possible to mount the semiconductor chip **30** more appropriately even when the size of the semiconductor chip **30** is relatively small.

(61) Moreover, as a comparison, there is a method of individually providing a plurality of spacer chips on sides of the semiconductor chip **20**. Examples of a material for the spacer chips include silicon (Si) and polyimide. At this time, a gap between the spacer chips and the semiconductor chip

20 is larger than the gap in the present embodiment. Due to this, even when the semiconductor chip **30** and the adhesive layer **40** are subsequently stacked, the gap cannot be buried only with the adhesive layer **60** flowing off from the adhesive layer **40**. Moreover, when the gap is subsequently buried with the encapsulation resin **91** and if heights of the spacer chips and the semiconductor chip **20** are small, flowability of the resin decreases to make it impossible to completely bury the gap with the encapsulation resin **91**. Alternatively, when the spacer chips are set higher for ensuring the flowability of the encapsulation resin **91**, it is impossible to realize miniaturization.

(62) Alternatively, when thinner spacer chips are used for ensuring the flowability of the encapsulation resin **91**, the gap between the semiconductor chip **20** and the spacer chips further widens. At this time, if the semiconductor chip **30** is thin and the semiconductor chip **30** is stacked on the spacer chips, the semiconductor chip **30** is prone to being warped and is sometimes broken due to the wide gap. Moreover, using the spacer chips increases cost and requirements, i.e., necessary conditions for a chip stacking device.

(63) According to the first embodiment, by contrast, the adhesive layer **60** buries the gap, i.e., through-hole **50h** between the semiconductor chip **20** and the spacer **50**. Therefore, the gap can be buried more easily than the case of using the encapsulation resin **91**. Furthermore, it is possible to make the semiconductor chip **30** and the spacer **50** thinner and further reduce a package height.

(64) It is noted that the spacer **50** is not limited to the resin layer. As the material for the spacer **50**, a material relatively high in rigidity for maintaining the height of the spacer **50** and easy to work for forming the through-hole **50h** may be used.

(65) Moreover, the height from the surface **F1** of the interconnection substrate **10** to the surface **50a** of the spacer **50** may be larger than the height from the surface **F1** of the interconnection substrate **10** to the surface **20a** of the semiconductor chip **20**. In this case, the heat is transferred from the interconnection substrate **10** to the adhesive layer **40** via the spacer **50** at the time of mounting the semiconductor chip **30**. In addition, the height of the semiconductor chip **20** is small, so that the adhesive layer **60** may also be provided between the upper surface of the semiconductor chip **20** and the adhesive layer **40**.

(66) Moreover, the spacer **50** has adhesion properties and may be directly adhesively bonded with the interconnection substrate **10**. In this case, the adhesive layer **51** may not be provided.

(67) Furthermore, in the processes of manufacturing the semiconductor device **1**, the spacer **50** (and the adhesive layer **51**) in which the marks **50m** and the through-hole **50h** are formed in advance may be used. In this case, it is possible to omit the processes of forming the marks **50m** and the through-hole **50h**.

(68) Furthermore, the adhesive layer **40** may be heated from not the interconnection substrate **10**-side but a semiconductor chip **30**-side. In this case, a heater is provided, for example, in a collet used to convey the semiconductor chip **30**. Furthermore, the heater may be provided in each of the interconnection substrate **10** and the collet.

Second Embodiment

(69) FIGS. **11** to **14** are cross-sectional views illustrating an example of a method of manufacturing the semiconductor device **1** according to a second embodiment.

(70) In the first embodiment, the through-hole **50h** is formed in the processes of preparing the adhesive layer **40**, the spacer **50**, and the adhesive layer **51** as one adhesive layer, as illustrated in FIGS. **2** to **4**. The adhesive layer **40**, the spacer **50**, and the adhesive layer **51** are subsequently bonded onto the semiconductor wafer **W** as one adhesive layer. In this case, the already formed through-hole **50h** is possibly unable to correspond to, for example, a change in specifications such as the sizes of the semiconductor chips **20** and **30**, possibly resulting in waste of the material. On the other hand, the second embodiment differs from the first embodiment in that the through-hole **50h** is formed in an assembly process.

(71) First, as illustrated in FIG. **11**, the adhesive layer **40** is provided on the dicing tape **DT**. Next, as illustrated in FIG. **12**, the semiconductor wafer **W** is made to adhere onto the adhesive layer **40**.

Next, as illustrated in FIG. 13, the dicing tape DT is peeled off from the semiconductor wafer W and the adhesive layer 40. The adhesive layer 40 is thereby provided on a lower surface, i.e., surface Wb that is one of surfaces of the semiconductor wafer W.

(72) Next, as illustrated in FIG. 14, the semiconductor wafer W is made to adhere onto the spacer 50 via the adhesive layer 40 provided on the surface Wb. More specifically, the adhesive layer 60 and the semiconductor wafer W are provided on the spacer 50 having the plurality of through-holes 50h in response to the positions of the plurality of through-holes 50h.

(73) The through-holes 50h are formed herein similarly to FIGS. 2 and 3 according to the first embodiment. It is noted that the through-holes 50h are formed in response to the size of at least one of the semiconductor chip 30 and the semiconductor chip 20 in an actual assembly process of the package (product). It is thereby possible to form the through-holes 50h in response to, for example, the change in specifications of the chip sizes or the like. As a result, it is possible to reduce waste of the materials for the spacer 50, the adhesive layer 51, and the like due to, for example, the change in specifications. The through-holes 50h may be formed before the process illustrated in FIG. 14 and may be formed, for example, in parallel to providing the adhesive layer 40 on the semiconductor wafer W.

(74) After the process illustrated in FIG. 14, the semiconductor device 1 illustrated in FIG. 1 is completed through similar processes to those illustrated in FIG. 6 and the following according to the first embodiment.

(75) The semiconductor device 1 according to the second embodiment can obtain similar effects to those of the first embodiment. Furthermore, according to the second embodiment, it is possible to form the through-holes 50h to correspond to the semiconductor chips 20 and 30 at the time of the actual assembly of the package (product) as described above.

(76) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A semiconductor device comprising: a substrate having a first surface; a first semiconductor chip disposed above the first surface; a first adhesive layer disposed on a lower surface of the first semiconductor chip, facing the substrate, the first adhesive layer containing a plurality of resins different in molecular weight from each other; a second semiconductor chip disposed between the substrate and the first adhesive layer; a second adhesive layer surrounding the second semiconductor chip as viewed from a normal direction of the first surface, the second adhesive layer containing at least one type of the resin lower in molecular weight than the other resins among the plurality of types of resins contained in the first adhesive layer; a spacer surrounding the second adhesive layer in the view from the normal direction of the first surface; and a sealing resin, different from the first adhesive layer and the second adhesive layer, covering the first semiconductor chip and the substrate; wherein the second adhesive layer only directly contacts a side surface of the second semiconductor chip.

2. The semiconductor device according to claim 1, wherein the second semiconductor chip is disposed such that the overall second semiconductor chip overlaps the first semiconductor chip as viewed from the normal direction of the first surface, and the second adhesive layer disposed such that the overall second adhesive layer overlaps the first adhesive layer in the view from the normal direction of the first surface.

3. The semiconductor device according to claim 1, wherein the spacer supports the first semiconductor chip and the first adhesive layer, and the spacer has a rigidity higher than a rigidity of the first adhesive layer.
 4. The semiconductor device according to claim 1, wherein the spacer is a resin layer.
 5. The semiconductor device according to claim 1, wherein the first adhesive layer and the second adhesive layer each include a filler, and a concentration of the filler in the second adhesive layer is lower than a concentration of the filler in the first adhesive layer.
 6. The semiconductor device according to claim 1, wherein from the first surface, a height of an upper surface of the spacer, the upper surface being opposed the first adhesive layer, is smaller than a height of an upper surface of the second semiconductor chip.
 7. The semiconductor device according to claim 1, wherein the substrate is an interconnection substrate.
 8. The semiconductor device according to claim 1, wherein the first semiconductor chip is a memory chip.
 9. The semiconductor device according to claim 8, wherein the first semiconductor chip is a NAND flash memory chip.
 10. The semiconductor device according to claim 8, wherein the second semiconductor chip is a memory controller chip.
 11. The semiconductor device according to claim 1, wherein the spacer is formed of polyimide.
 12. The semiconductor device according to claim 1, wherein the first adhesive layer contains a high-molecular-weight base material and a low-molecular-weight epoxy resin relative to the base material.
 13. The semiconductor device according to claim 1, wherein the first adhesive layer has a thickness equal to or larger than one half a thickness of the second semiconductor chip.
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